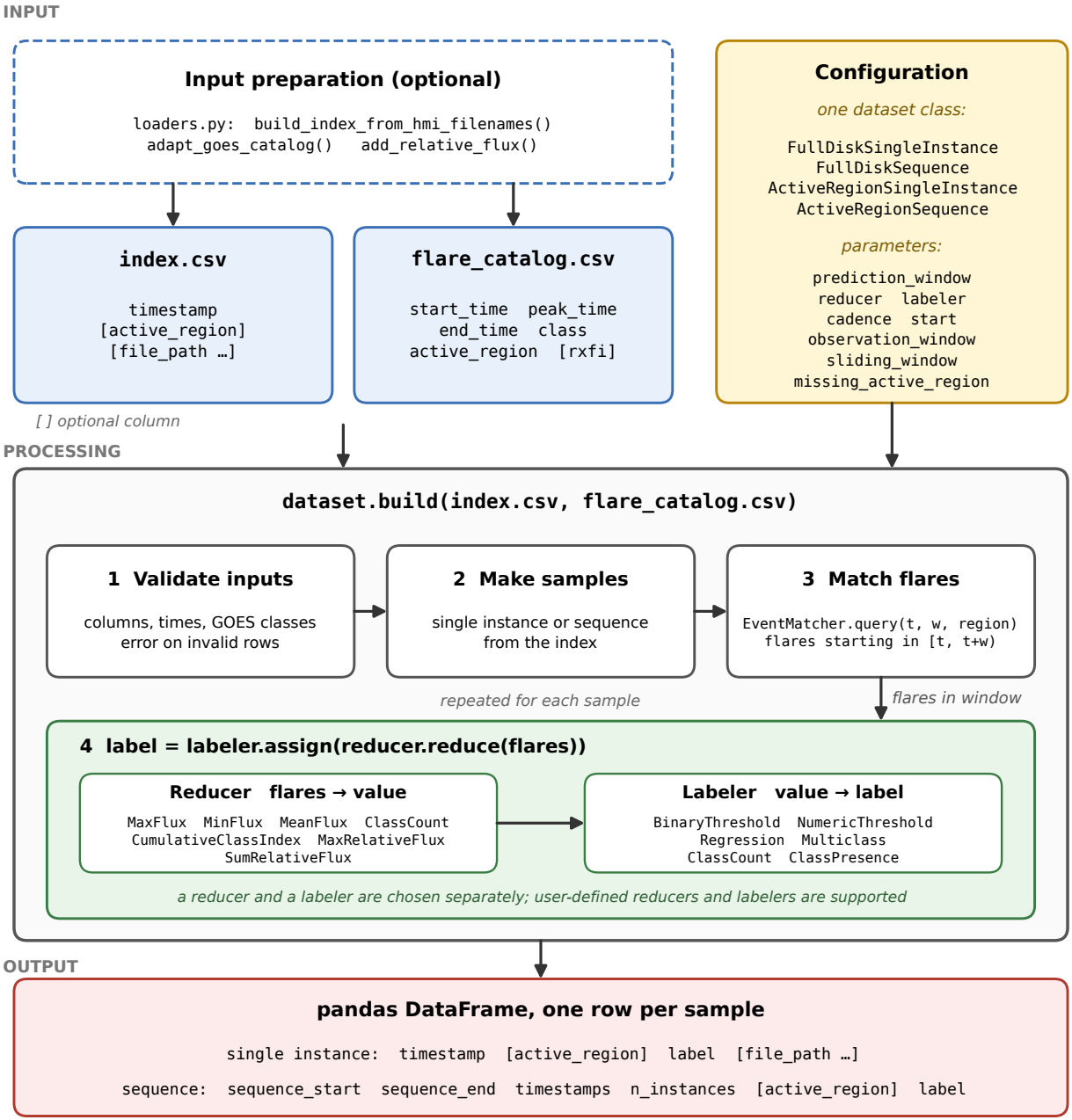
INPUT
Input preparation (optional)
loaders.py: build_index_from_hmi_filenames()
adapt_goes_catalog() add_relative_flux()
index.csv
timestamp
[active_region]
[file_path …]
flare_catalog.csv
start_time peak_time
end_time class
active_region [rxfi]
Configuration
one dataset class:
FullDiskSingleInstance
FullDiskSequence
ActiveRegionSingleInstance
ActiveRegionSequence
parameters:
prediction_window
reducer labeler
cadence start
observation_window
sliding_window
missing_active_region
[ ] optional column
PROCESSING
dataset.build(index.csv, flare_catalog.csv)
1 Validate inputs
columns, times, GOES classes
error on invalid rows
2 Make samples
single instance or sequence
from the index
3 Match flares
EventMatcher.query(t, w, region)
flares starting in [t, t+w)
repeated for each sample
flares in window
4 label = labeler.assign(reducer.reduce(flares))
Reducer flares → value
MaxFlux MinFlux MeanFlux ClassCount
CumulativeClassIndex MaxRelativeFlux
SumRelativeFlux
Labeler value → label
BinaryThreshold NumericThreshold
Regression Multiclass
ClassCount ClassPresence
a reducer and a labeler are chosen separately; user-defined reducers and labelers are supported
OUTPUT
pandas DataFrame, one row per sample
single instance: timestamp [active_region] label [file_path …]
sequence: sequence_start sequence_end timestamps n_instances [active_region] label